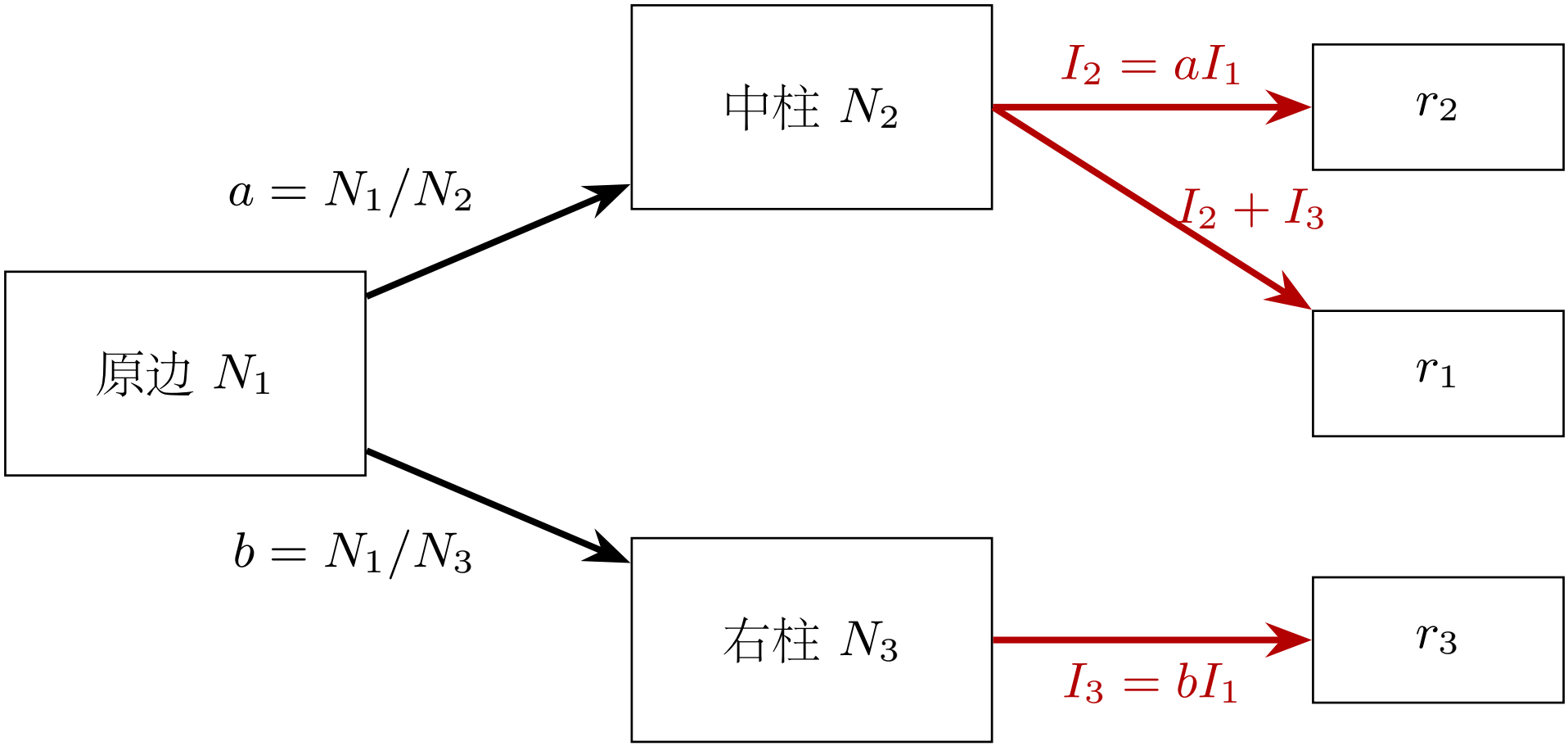

$$D = r_1' + a^2 r_2 + b^2 r_3 + (a+b)^2 r_1$$

$$P_2 = \frac{1}{2} r_2 a^2 \left(\frac{U_0}{D} \right)^2$$